AIR POLLUTION

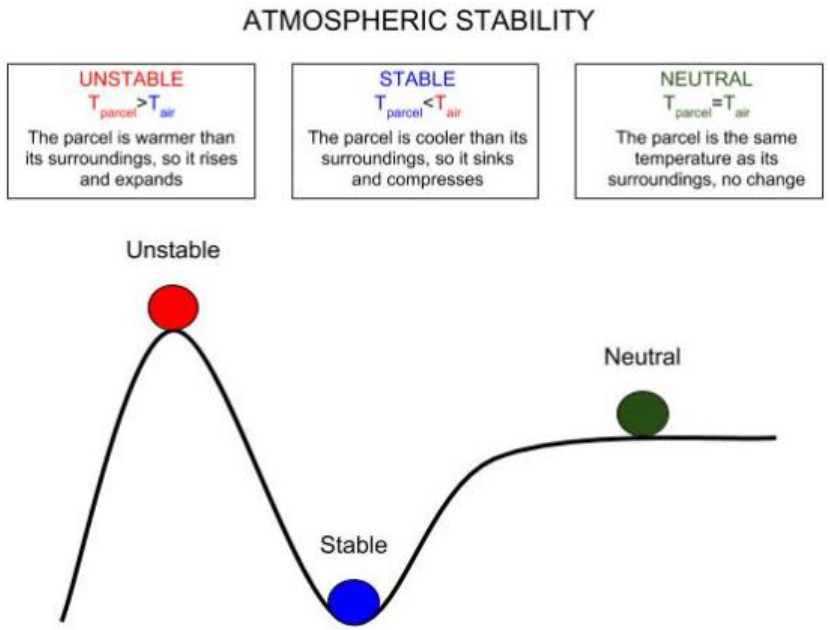
Debasree Purkayastha
School of Infrastructure, IIT Bhubaneswar
Email id: debasreep@iitbbs.ac.in

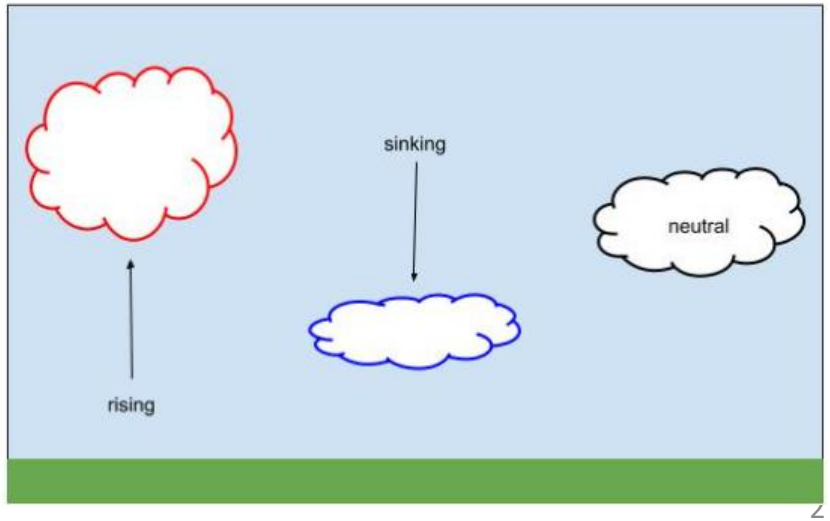
Atmospheric Stability

- If an air parcel is warmer than its surrounding environment, then it will be less dense than its surroundings and will rise like a hot air balloon. This is **Unstable Air** and has the potential for creating storms.

- If an air parcel is cooler than its surrounding environment, then it will be denser than its environment and will sink. Think of this situation as a pebble sinking in water. This is **Stable Air** which generally leads to clear skies.

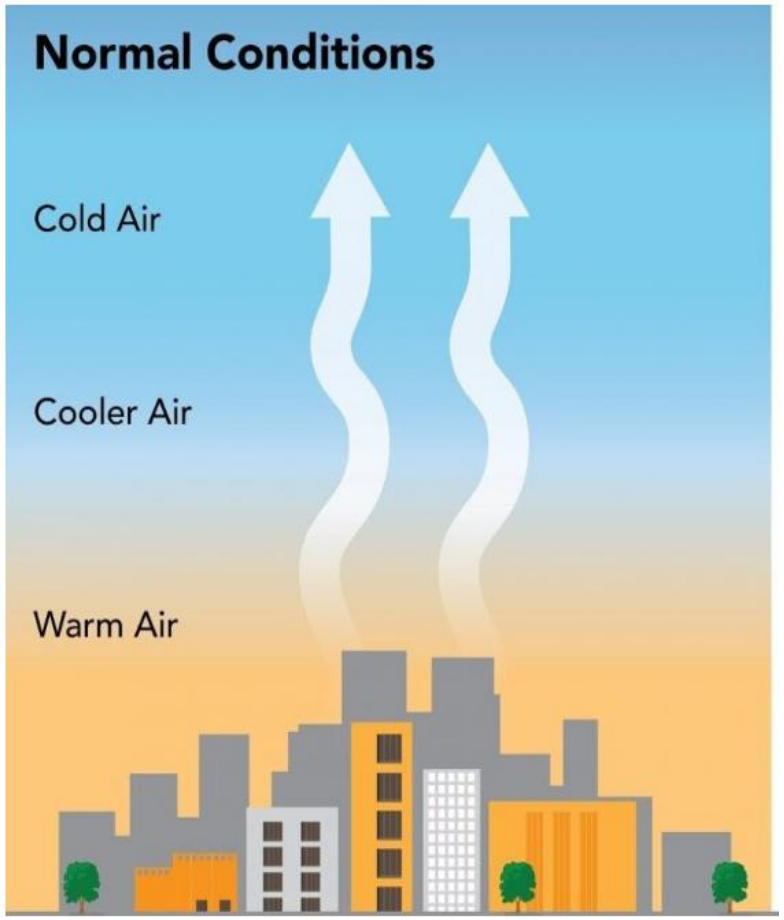
- If an air parcel is the same temperature as its surrounding environment, then the parcel will not move of its own accord. This is **Neutral Air**.





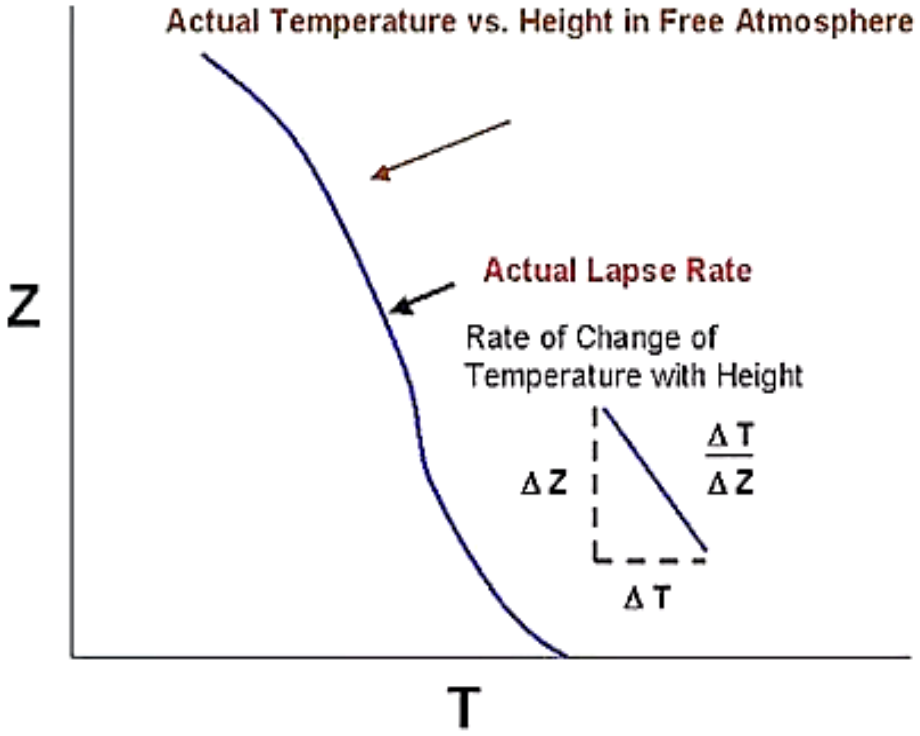
Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985)

Lapse Rate



- *Lapse rate is the rate at which temperature decreases as the altitude increases.*

$$Lapse\ Rate = -\frac{dT}{dA}$$

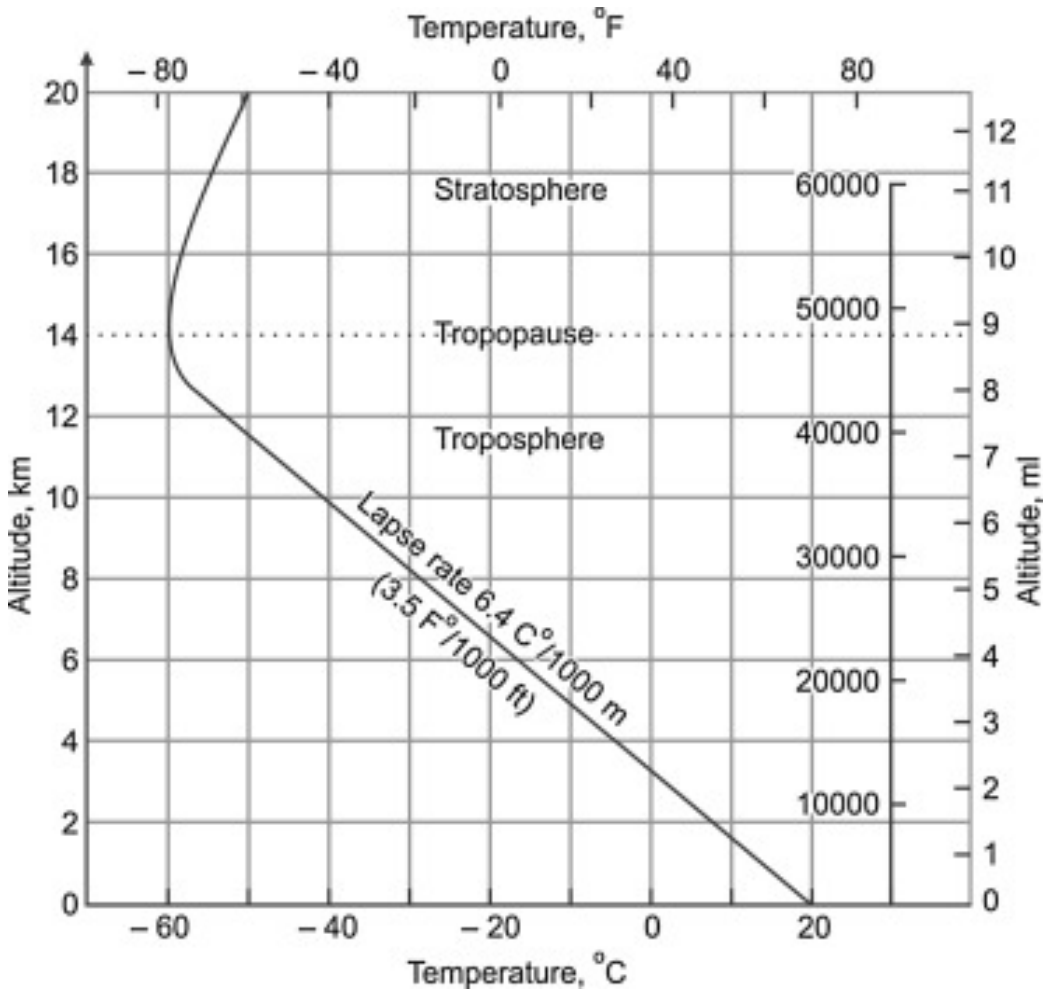


Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Image Source: https://lotusarise.com/temperature-inversion-upsc/

Environmental Lapse Rate

- In the troposphere, the temperature of the ambient air usually decreases with an increase in altitude.
- This rate can be determined for a particular place at a particular time by sending up a balloon equipped with a thermometer.

- The balloon moves through the air, not with it, and the temperature gradient of ambient air, which the rising balloon measures, is called the ambient lapse rate, the **environmental lapse rate**, or the prevailing lapse rate.



Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Image: https://www.sciencedirect.com/science/article/abs/pii/B9780128119891000142

Adiabatic Lapse Rate

- Change of temperature with a change in altitude of an air parcel without gaining or losing any heat to the environment surrounding the parcel is called the Adiabatic Lapse Rate (ALR).

- Assumptions for an Air Parcel

  i. An air parcel is thermally insulated from the surroundings

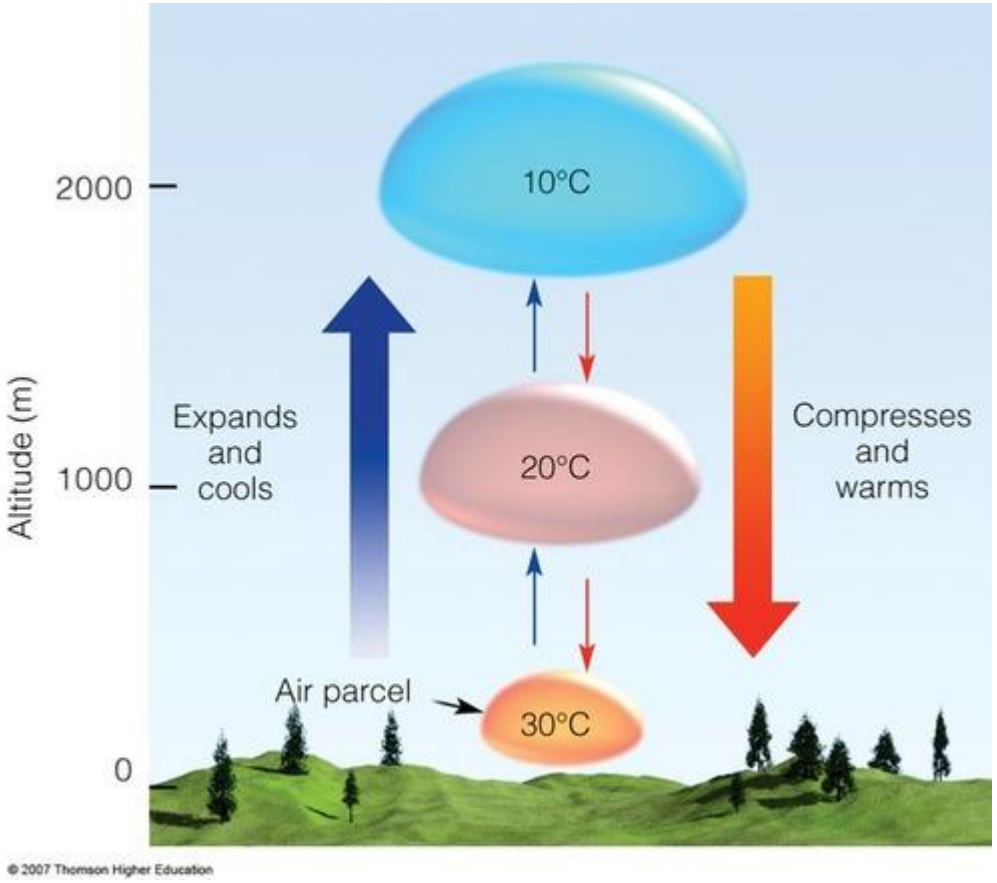
  ii. Air parcel is at the same pressure as the environmental air at the same level

  iii. The kinetic energy of the process is a negligible fraction of its total energy. This implies that the parcel moves very slowly.

- Two types of ALR:

  A. Dry Adiabatic Lapse Rate

  B. Wet Adiabatic Lapse Rate

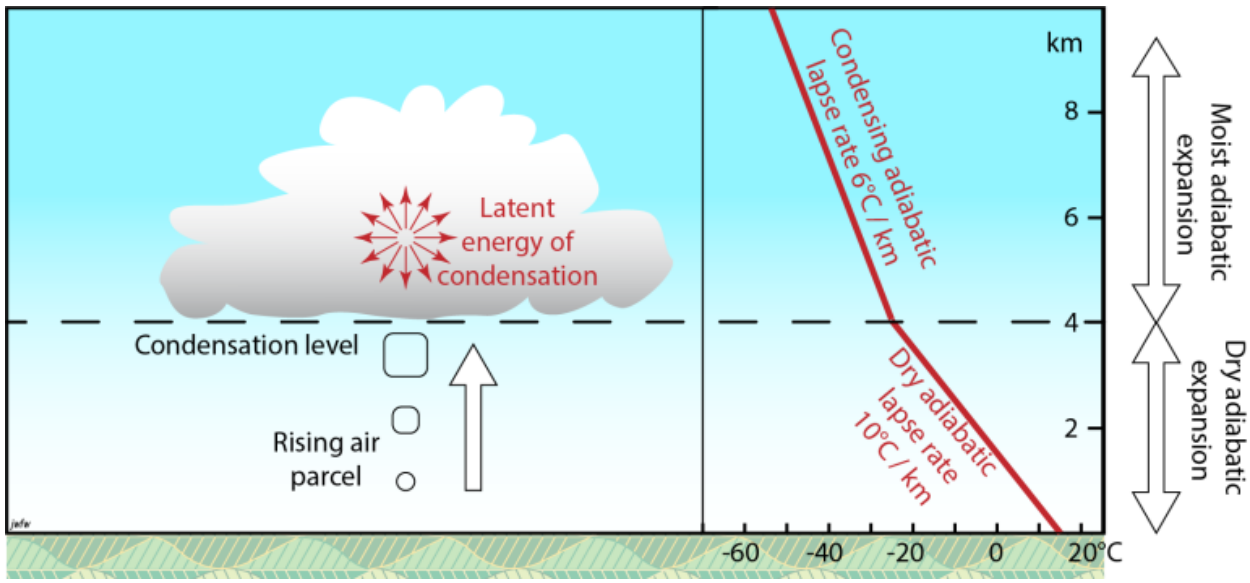


Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Image: https://www.epa.gov/expobox/exposure-assessment-tools-media-air

Adiabatic Lapse Rate

- The Dry Adiabatic Lapse Rate (DALR) is the rate at which the temperature of a dry (unsaturated) air parcel decreases as it rises in the atmosphere, approximately 9.8°C per kilometer (or 5.5°F per 1000 feet).

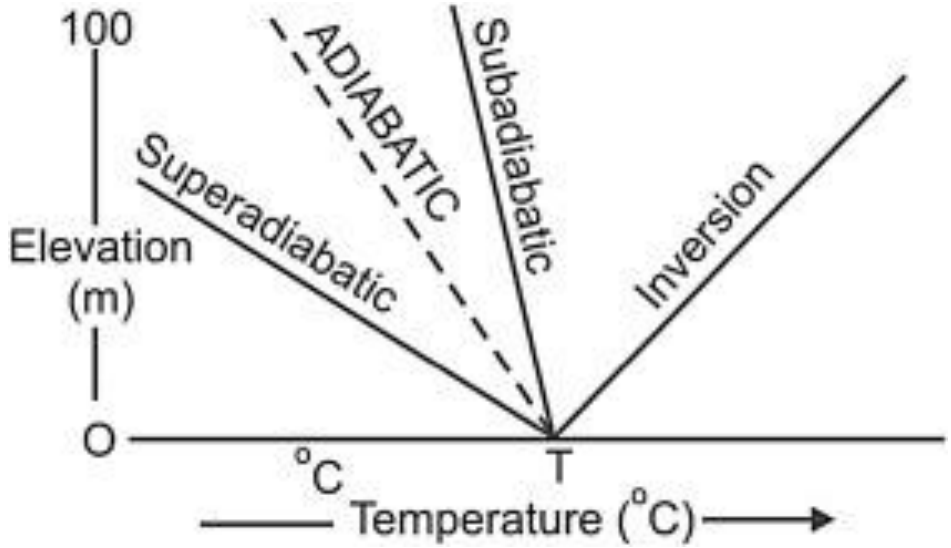
- As parcel rises, H_2O condenses and gives off heat, and warms air around it. Parcel cools more slowly as it rises in altitude, ≈6°C/1000 m (≈3°F/1000 ft). This lapse rate is called Wet Adiabatic Lapse Rate.



Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Adiabatic lapse rate, https://www.sciencedirect.com/topics/earth-and-planetary-sciences/adiabatic-lapse-rate, 3. Image Source: https://pressbooks.openeducationalberta.ca/planetearth/chapter/the-atmosphere/

Superadiabatic Lapse Rate

- **Superadiabatic:** ELR > ALR indicates unstable atmosphere. Vertical motion and mixing processes are enhanced. Dispersion of pollution plume is enhanced.
- An air parcel going up will continue to go up as the air parcel becomes warmer.
- An air parcel going down will continue to go down as the parcel becomes colder than the surrounding air.
- This condition can be compared with a 'ball on the concave down surface'. The ball on the surface tends to move in either direction.



Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985)

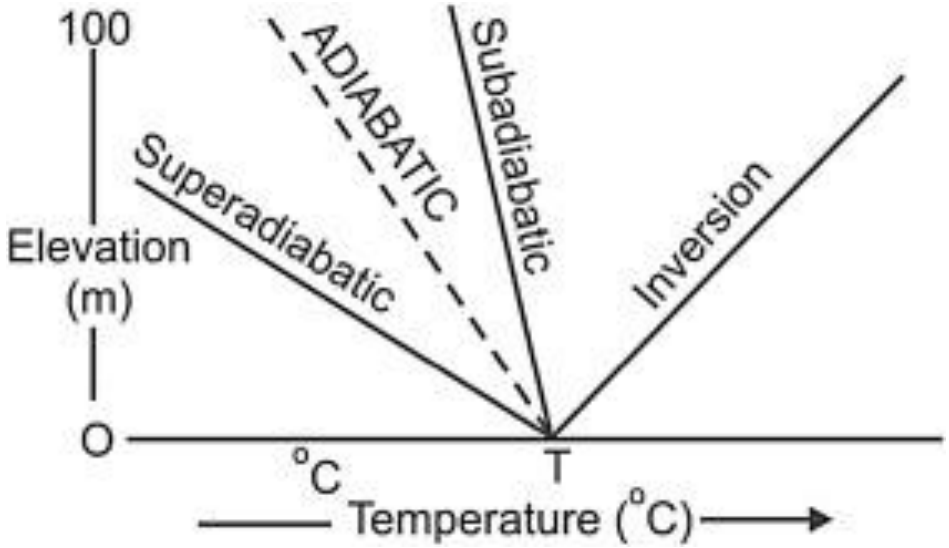
Subadiabatic Lapse Rate

- **Subadiabatic:** ELR < ALR . It indicates stable atmosphere, vertical motion, and mixing are suppressed. Dispersion is suppressed, and contamination is trapped.

- If the parcel of air rises up, it becomes colder than the surrounding air. Hence, comes down to the original position.

- If the parcel of air goes down, it becomes warmer than the surrounding air. Hence, rises up to the original position.

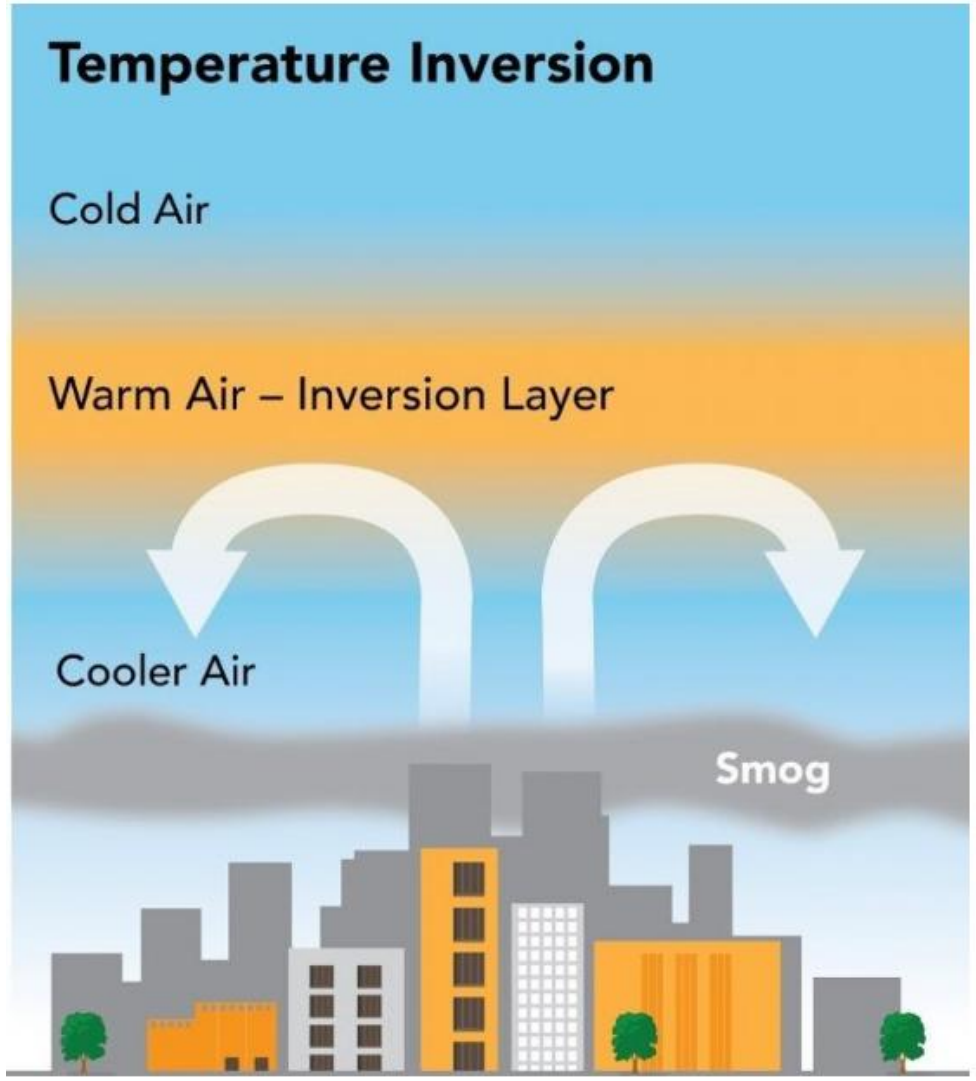
- This condition can be compared with a 'ball on the concave upward surface'. The ball on the surface tends to come back to the central original position.



Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985); 2. Adiabatic lapse rate, https://www.sciencedirect.com/topics/earth-and-planetary-sciences/adiabatic-lapse-rate

Inversion

- **Inversion:** An extreme case of subadiabatic, where temperature actually increases with altitude near the ground before it begins to decrease with altitude. This results in warm, low-density air riding on top of cool high density air; a very stable air column that traps pollution near the ground, like that which occurs during the winter in Delhi



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Numerical

Following observations have been made for the temperature of the atmosphere with respect to elevation to find the stability of the atmosphere. Determine the stability of the atmosphere.

Elevation(in m)	Temperature (in degree C)
10	15
40	14.5
100	13.5

- Solution

$\Delta T/\Delta Z$ = (15-14.5)/(40-10) = 0.5/30 = 16.7 degree C per km

$\Delta T/\Delta Z$ = (14.5-13.5)/(100-40) = 6/40 = 16.7 degree C per km

So, ELR = 16.7 degree C per km

We know, ALR = 9.8 degree C per km

ELR > ALR (super-adiabatic condition)

So, the atmosphere is unstable.

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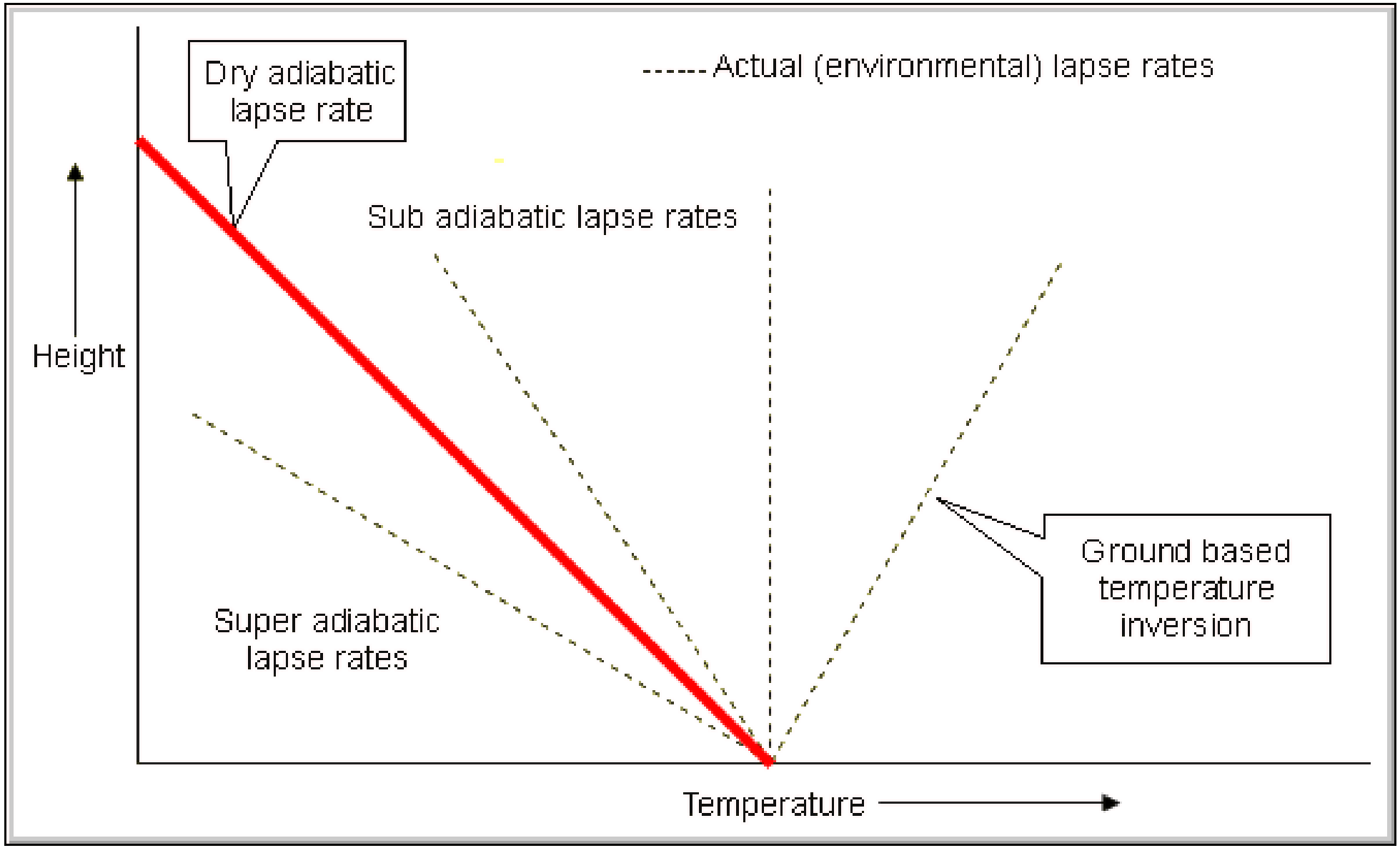
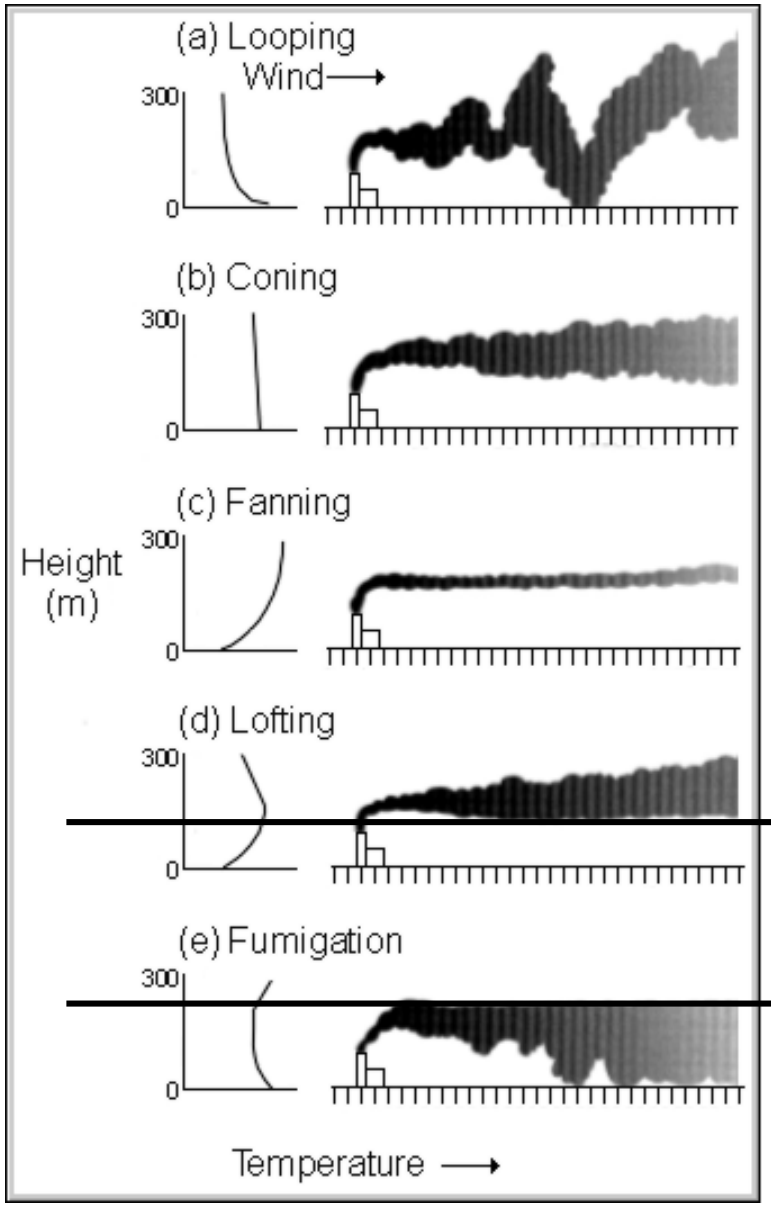

Dry adiabatic lapse rate
------ Actual (environmental) lapse rates
Sub adiabatic lapse rates
Height
Super adiabatic lapse rates
Ground based temperature inversion
Temperature

Pollution dispersion patterns from a stack



Super adiabatic conditions

Sub adiabatic conditions, neutral conditions

Stable conditions, very little mixing

Discharge above the inversion layer

a super-adiabatic lapse rate above an inversion.

Discharge below the inversion layer

a super-adiabatic lapse rate beneath an inversion.

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Image Source: https://vula.uct.ac.za/access/content/group/c7716642-ada7-42a0-acee-0434c085df90/electronic%20resources/cd3/env/air-pollution/APChapter5-1.htm

Air pollution Control approaches

- **Dilution:** Tall stacks may penetrate the inversion layer and disperse the contaminants, so that ground level contaminants are greatly reduced.

- **Control at Source:** Preventing the contaminant from coming by substituting alternate power source (such as geothermal, solar, hydraulic, etc.), Proper use of existing equipment by proper operation and maintenance of equipment, Changing the process being used (for e.g. instead of using open-hearth furnaces using controlled oxygen furnaces or electric furnaces).

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Air pollution Control Devices

Control devices can be divided into five major groups:

- Gravitational Settling Chamber
- Centrifugal Collectors
- Wet Collectors
- Electrostatic Precipitators
- Fabric filters

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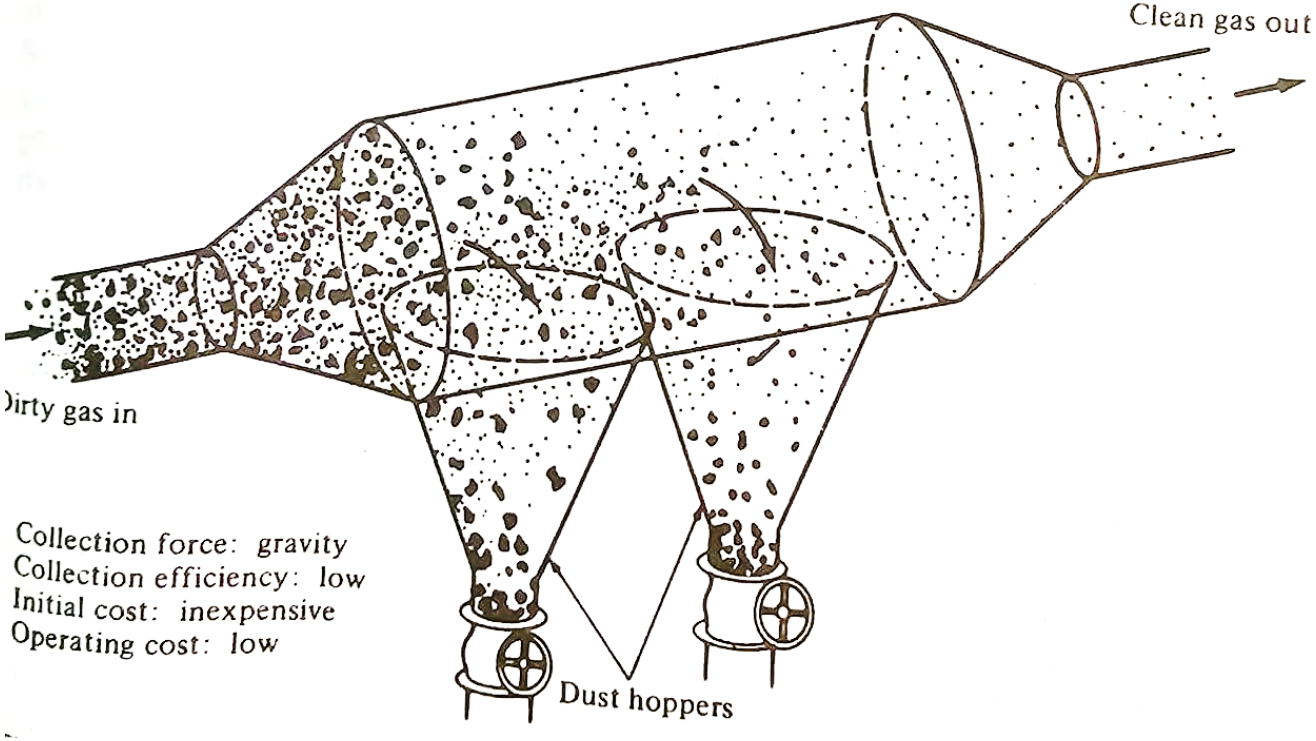
Gravitational Settling Chamber

- Settling chambers in the air pollution control system provides enlarged areas to minimize horizontal velocities and allow time for the vertical velocity to carry the particle to the floor.



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Assignment

Sketch the schematic and write the working principle of following control devices:

- Electrostatic Precipitators
- Fabric filters

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Thank you

For any questions, please feel free to reach out to me.